

WBA9 – Butte City groundwater storage: modeled outcome levels

A summary brief reporting modeled outcome levels at one location of interest under every management strategy in the library, read across the five hydroclimates. Each difference from the baseline is split into the hydroclimate effect and the management effect. Run identifiers for every scenario are listed in the references note.

Summary. This brief reports modeled groundwater storage outcome levels at WBA9 - Butte City under each of the 28 management strategies and the five hydroclimates in the COEQWAL library. Figure 1 places every result beside the baseline, current operations under the historical hydroclimate, and splits each difference into the hydroclimate effect and the management effect. The table beneath the figure gives the same values as numbers, and the companion data file holds all of them. Cells outlined in the table are those where the two effects run opposite ways; the largest such offset is *Groundwater pumping limits in the Central Valley* under extreme climate stress, where the hydroclimate effect is +1.0 and the management effect is -0.8.

What this is

This brief reports modeled groundwater storage outcome levels for WBA9 - Butte City, one of the 42 groundwater basin areas COEQWAL tracks as a location of interest (LOI). It covers all 28 strategies across all five modeled hydroclimates in the COEQWAL library. Every LOI has a brief of this form, built around that location's results. Where a key outcome includes more than one location, Figure 2 also shows the other locations' results alongside this one's. A scenario is one strategy under one hydroclimate; the five hydroclimates are a sample of stress levels, not a forecast, and the hydroclimates reference brief describes how they were built.

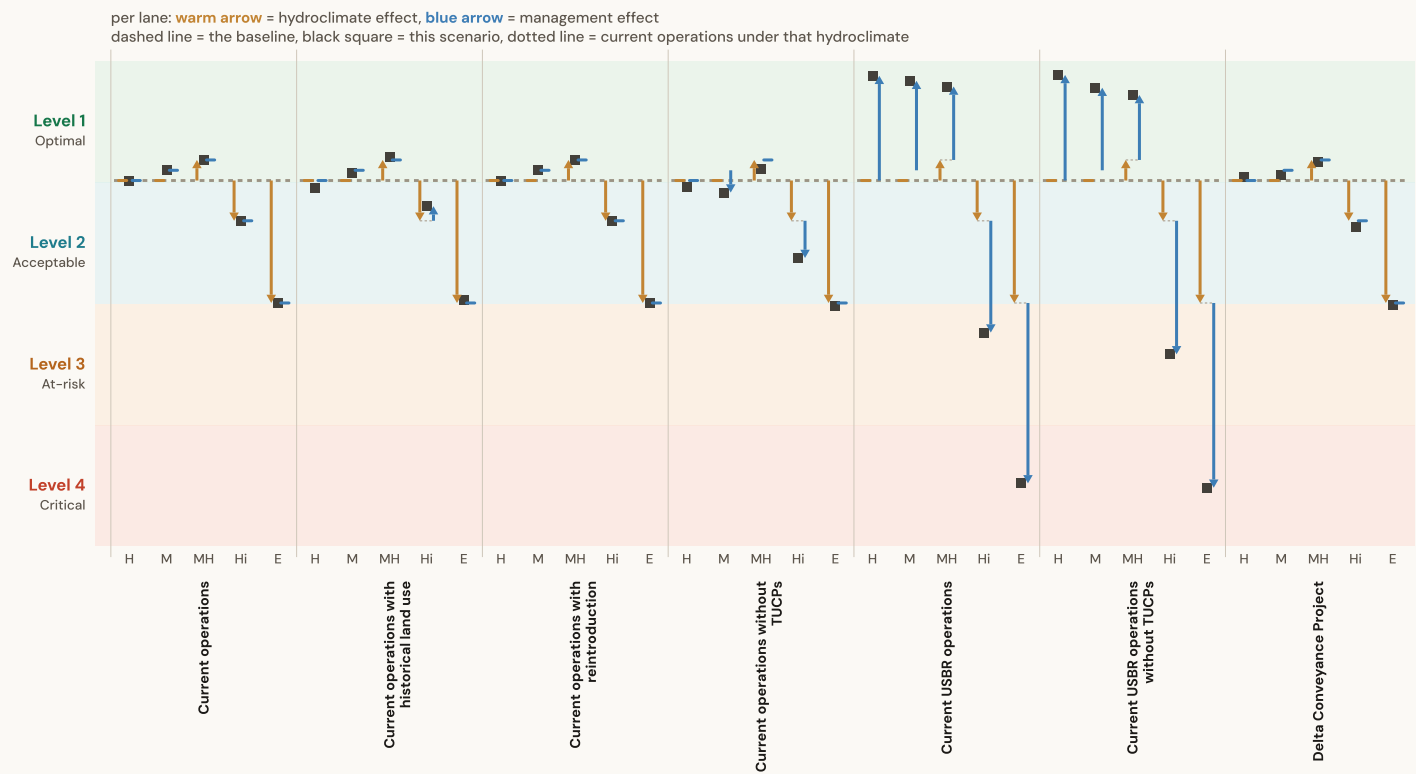
How to read the figure

Each mark in Figure 1 is the modeled outcome level at WBA9 - Butte City under one strategy and one hydroclimate. The baseline value at this location is 1.9, and it is the same reference in every column. Each difference from the baseline is drawn as two arrows. The warm arrow is the hydroclimate effect: how much current operations under that hydroclimate differ from the baseline. The blue arrow is the management effect: how much the strategy differs from current operations under the same hydroclimate. The two add to the difference from the baseline; a flat tick is a change smaller than 0.1.

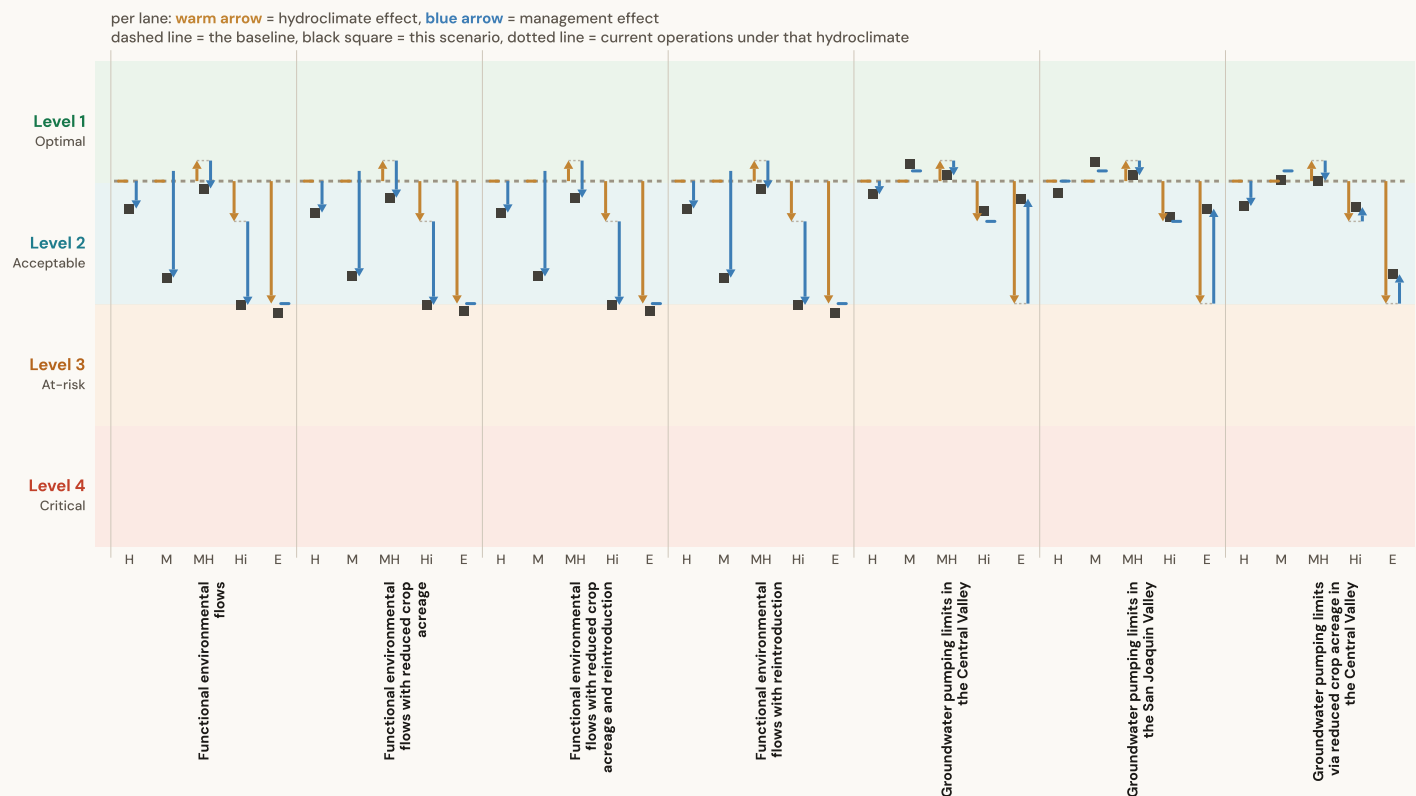
Values sit on a continuous axis from 1 to 4.99 with Level 1 (Optimal) at the top, and the shaded bands are the four levels. A positive effect moves a value toward Level 4 (Critical); a negative effect moves it toward Level 1 (Optimal). Every value here is this location's own; nothing on this page is a mean, and Figure 2 places the location among the 42 groundwater storage locations. How the split works is explained in [Separating the effects of strategy and hydroclimate in COEQWAL results](#); the scale, thresholds, and rounding rules are in [How to read the values below](#).

All 28 strategies, all five hydroclimates

Panel 1 of 4 — strategies 1 to 7 of 28

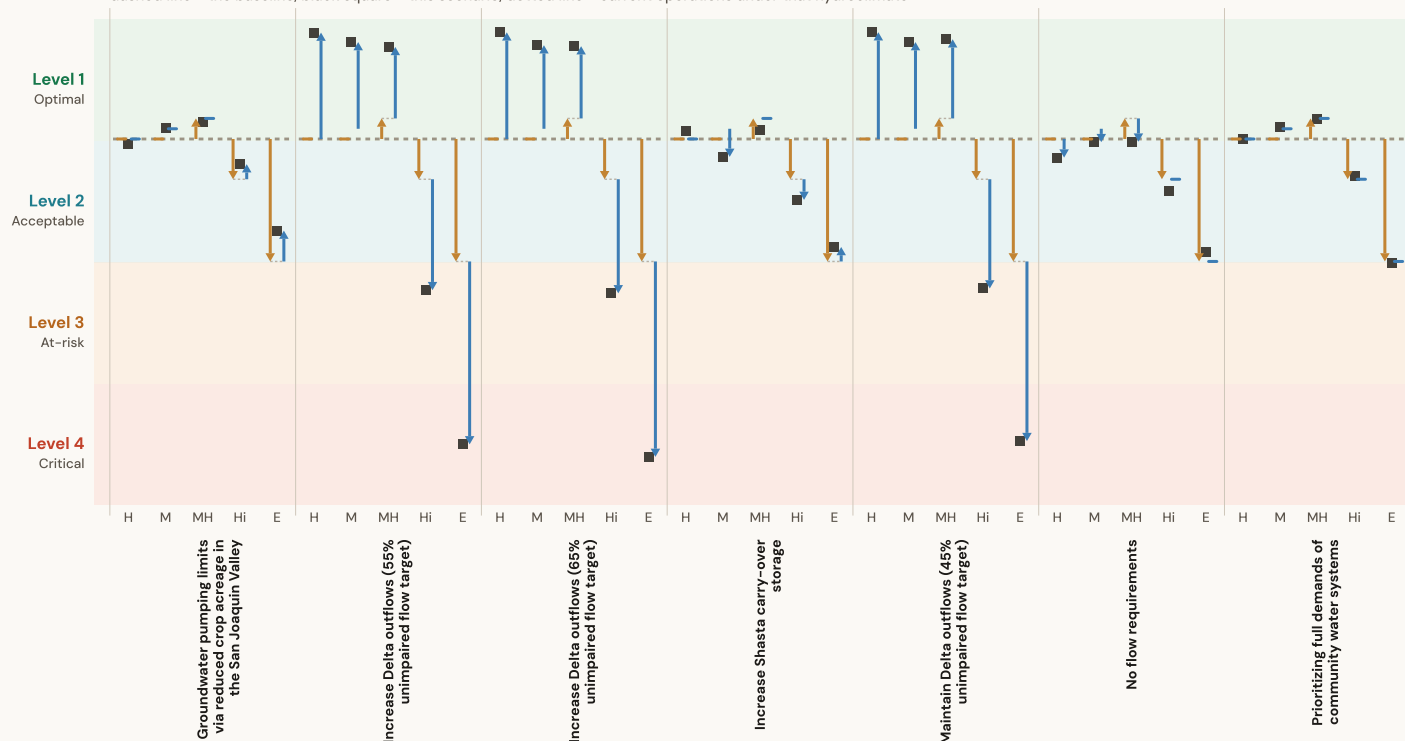


Panel 2 of 4 — strategies 8 to 14 of 28



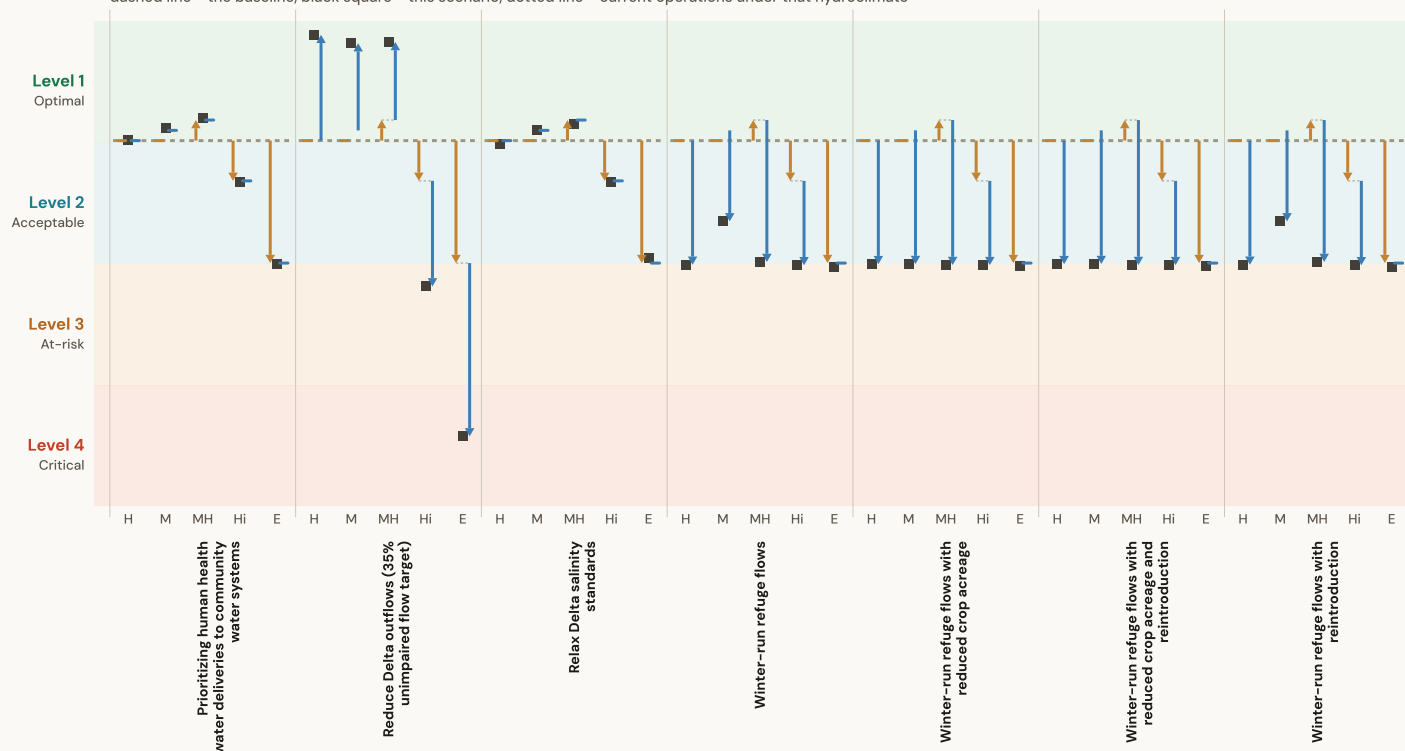
Panel 3 of 4 — strategies 15 to 21 of 28

per lane: **warm arrow** = hydroclimate effect, **blue arrow** = management effect
dashed line = the baseline, black square = this scenario, dotted line = current operations under that hydroclimate



Panel 4 of 4 — strategies 22 to 28 of 28

per lane: **warm arrow** = hydroclimate effect, **blue arrow** = management effect
dashed line = the baseline, black square = this scenario, dotted line = current operations under that hydroclimate



The 28 strategies are shown in four stacked panels that share one level axis. The strategies are split across four panels only to fit the page; read them as one figure.

Figure 1. Modeled level values for WBA9 - Butte City and the two effects that produce their difference from the baseline, 28 management strategies × five hydroclimates. Lanes within each strategy run left to right: Historical (H), Moderate (M), Moderate-high (MH), High (Hi),

Extreme (E) climate stress. Vertical axis: continuous level value, Level 1 at top, level bands as shaded zones. Dashed level: the baseline. Black square: this scenario. A short dotted line where the warm arrow ends and the blue arrow begins marks current operations under that hydroclimate. A short flat tick in place of an arrow is a change smaller than 0.1 on the level scale. Every mark is a square, which is this set's mark for a value that is one location's rather than a mean over several. Every value is in the table that follows, and the scale is set out in how to read the values.



Figure 2. Groundwater storage results, presented as modeled outcome levels, across the 42 groundwater storage locations, with WBA9 - Butte City marked. Each column shows the span of modeled results for one groundwater basin area across the COEQWAL library. Ends of a column are the lowest and the highest modeled value at that location across the runs that report it. A short dash across a column is that location's own baseline (current operations under the historical hydroclimate). This figure uses the same value axis (y-axis) format as Figure 1. Every column uses the same encoding; the column for WBA9 - Butte City, this brief's subject, is drawn wider, in a distinct color, and marked above. The scale is set out in how to read the values.

The same 140 results as numbers

Each cell reads top to bottom: the hydroclimate effect, the management effect, and the resulting value with its band. The two effects add to the difference between the scenario and the baseline. Outlined cells are those where the two run opposite ways. Readers who want to cite a specific column can link to it directly, and the same numbers appear in the companion data file.

Strategy		The baseline	Historical	Moderate	Moderate-high	High	Extreme	Management effect across the five
Current operations SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.9	L1 1.9	L1 1.8	L2 2.3	L2 2.9	no change at the 0.1 threshold 0.0
Current operations with historical land use SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 2.0	L1 1.9	L1 1.7	L2 2.1	L2 2.9	one direction -0.2 to 0.0
Current operations with reintroduction SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.9	L1 1.9	L1 1.8	L2 2.3	L2 2.9	no change at the 0.1 threshold 0.0
Current operations without TUCPs SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 2.0	L2 2.0	L1 1.8	L2 2.6	L3 3.0	one direction 0.0 to +0.3
Current USBR operations SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.1	L1 1.1	L1 1.2	L3 3.2	L4 4.4	crosses zero -0.8 to +1.5
Current USBR operations without TUCPs SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.1	L1 1.2	L1 1.2	L3 3.4	L4 4.5	crosses zero -0.8 to +1.6
Delta Conveyance Project SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.9	L1 1.9	L1 1.8	L2 2.3	L3 3.0	no change at the 0.1 threshold 0.0
Functional environmental flows SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 2.2	L2 2.7	L2 2.0	L3 3.0	L3 3.0	one direction 0.0 to +0.8
Functional environmental flows with reduced crop	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 2.2	L2 2.7	L2 2.1	L3 3.0	L3 3.0	one direction 0.0 to +0.8

Strategy		The baseline	Historical	Moderate	Moderate-high	High	Extreme	Management effect across the five
acreage SINGLE LEVEL VALUE								
Functional environmental flows with reduced crop acreage and reintroduction SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 0.0 +0.3 2.2	L2 0.0 +0.8 2.7	L2 -0.1 +0.3 2.1	L3 +0.4 +0.7 3.0	L3 +1.0 0.0 3.0	one direction 0.0 to +0.8
Functional environmental flows with reintroduction SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 0.0 +0.3 2.2	L2 0.0 +0.8 2.7	L2 -0.1 +0.2 2.0	L3 +0.4 +0.7 3.0	L3 +1.0 0.0 3.0	one direction 0.0 to +0.8
Groundwater pumping limits in the Central Valley SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 0.0 +0.1 2.0	L1 0.0 0.0 1.8	L1 -0.1 +0.1 1.9	L2 +0.4 0.0 2.2	L2 +1.0 -0.8 2.1	crosses zero -0.8 to +0.1
Groundwater pumping limits in the San Joaquin Valley SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 0.0 0.0 2.0	L1 0.0 0.0 1.8	L1 -0.1 +0.1 1.9	L2 +0.4 0.0 2.2	L2 +1.0 -0.7 2.2	crosses zero -0.7 to +0.1
Groundwater pumping limits via reduced crop acreage in the Central Valley SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 0.0 +0.2 2.1	L1 0.0 0.0 1.9	L1 -0.1 +0.1 1.9	L2 +0.4 -0.2 2.1	L2 +1.0 -0.2 2.7	crosses zero -0.2 to +0.2
Groundwater pumping limits via reduced crop acreage in the San Joaquin Valley SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 0.0 0.0 2.0	L1 0.0 0.0 1.8	L1 -0.1 0.0 1.8	L2 +0.4 -0.2 2.1	L2 +1.0 -0.2 2.7	one direction -0.2 to 0.0

		The baseline		Historical		Moderate		Moderate-high		High		Extreme		Management effect across the five
Strategy														
Increase Delta outflows (55% unimpaired flow target) SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.1	L1 1.1	L1 1.1	L1 1.2	L3 3.2	L4 4.4	crosses zero -0.8 to +1.5					
Increase Delta outflows (65% unimpaired flow target) SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.1	L1 1.2	L1 1.2	L3 3.2	L4 4.5	crosses zero -0.8 to +1.6						
Increase Shasta carry-over storage SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.9	L2 2.1	L1 1.9	L2 2.4	L2 2.8	crosses zero -0.1 to +0.2						
Maintain Delta outflows (45% unimpaired flow target) SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.1	L1 1.1	L1 1.1	L3 3.2	L4 4.4	crosses zero -0.8 to +1.5						
No flow requirements SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L2 2.1	L2 2.0	L2 2.0	L2 2.4	L2 2.9	one direction 0.0 to +0.2						
Prioritizing full demands of community water systems SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.9	L1 1.8	L1 1.8	L2 2.2	L3 3.0	no change at the 0.1 threshold 0.0						
Prioritizing human health water deliveries to community water systems SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.9	L1 1.8	L1 1.8	L2 2.3	L3 3.0	no change at the 0.1 threshold 0.0						
Reduce Delta outflows (35% unimpaired flow target) SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	L1 1.1	L1 1.1	L1 1.1	L3 3.1	L4 4.4	crosses zero -0.8 to +1.5						

Strategy		The baseline	Historical	Moderate	Moderate-high	High	Extreme	Management effect across the five
Relax Delta salinity standards SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	0.0 0.0 L2 2.0	0.0 0.0 L1 1.8	-0.1 0.0 L1 1.8	+0.4 0.0 L2 2.3	+1.0 0.0 L2 2.9	no change at the 0.1 threshold 0.0
Winter-run refuge flows SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	0.0 +1.1 L3 3.0	0.0 +0.7 L2 2.6	-0.1 +1.1 L2 2.9	+0.4 +0.7 L3 3.0	+1.0 0.0 L3 3.0	one direction 0.0 to +1.1
Winter-run refuge flows with reduced crop acreage SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	0.0 +1.1 L3 3.0	0.0 +1.1 L3 3.0	-0.1 +1.2 L3 3.0	+0.4 +0.7 L3 3.0	+1.0 0.0 L3 3.0	one direction 0.0 to +1.2
Winter-run refuge flows with reduced crop acreage and reintroduction SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	0.0 +1.1 L3 3.0	0.0 +1.1 L3 3.0	-0.1 +1.2 L3 3.0	+0.4 +0.7 L3 3.0	+1.0 0.0 L3 3.0	one direction 0.0 to +1.2
Winter-run refuge flows with reintroduction SINGLE LEVEL VALUE	Hydroclimate effect Management effect Value (level)	L1 1.9	0.0 +1.1 L3 3.0	0.0 +0.7 L2 2.6	-0.1 +1.1 L2 2.9	+0.4 +0.7 L3 3.0	+1.0 0.0 L3 3.0	one direction 0.0 to +1.1

About this location

Reported values	Modeled outcome levels, Level 1 (Optimal) to Level 4 (Critical) — see How to read the values
Underlying model variable	Groundwater storage in this water budget area, from CalSim 3
Type	Wba
Key outcome	Groundwater storage
A description of this location has not yet been added.	

Related briefs

Briefs for all locations of interest are available from the COEQWAL Data page.

How were the five hydroclimates constructed? The hydroclimates reference brief defines them. It reports no results of its own; every brief above is read against the same five.

What this doesn't say

This brief reports modeled results from the COEQWAL ensemble. It does not predict what will happen, evaluate whether those results are acceptable, or recommend any course of action. The outcome-level framework provides definitional categories; the meaning of those categories for any individual reader, community, or decision is not specified by COEQWAL.

In particular, for this brief: the five hydroclimates are a sample of stress levels, carry no probabilities, and are not a timeline, so no average over them is reported here. Splitting a difference into two effects is arithmetic on modeled values, not a causal claim about the water system, and the two are separable only because each is defined against its own reference. A step or a spread on the level axis is a modeled magnitude, not a benefit or a cost. This page reports one modeled location, so nothing here is a mean and nothing here describes the key outcome the location sits in. The scale has ends: a location whose baseline sits in Level 1 can only shift toward higher-numbered levels, and one in Level 4 only toward lower-numbered ones. Differences in how much locations shift partly reflect where each starts. Locations are modeled nodes, not measured conditions on the ground.

A strategy's highest value does not necessarily fall under extreme climate stress.

References, run identifiers and the build stamp

Model runs (current operations): Historical `s0020`; Moderate `s0134`; Moderate-high `s0047`; High `s0056`; Extreme `s0108`. Each row reads its own five runs beside these; they are in the companion data file at `#/rows/*/by_hydroclimate/*/run_ids`. The baseline is current operations under the historical hydroclimate, `s0020`.

Build stamp. Snapshot `2026-08-29`; builder version `0.1.0`; built 2026-09-25T20:54:30Z; snapshot data read 2026-08-29; verification: pass. The verification status is written by the independent check (`briefbuilder.verifier`), which recomputes every value in the companion data file from the per-location data and reads every arrow on this page, after both files are written. It is not stamped by whatever produced them.

Companion data file: `loi_brief_09_schema1.json`, holding every value in this brief plus the per-location counts, the spread measures, the derivation of every number in the prose and the figure's declared attribute contract. Per-hydroclimate slices: Historical `#/rows/*/by_hydroclimate/BL`; Moderate `#/rows/*/by_hydroclimate/HC4`; Moderate-high `#/rows/*/by_hydroclimate/HC1`; High `#/rows/*/by_hydroclimate/HC2`; Extreme `#/rows/*/by_hydroclimate/HC3`. Built from the locked 2026-08-29 API snapshot (api.coeqwal.org v2.1.1 (locked dataset snapshot)). Decomposition per `Decomposition_Math_Quickref` (coeqwal-viz): the hydroclimate effect is current operations under that hydroclimate minus current operations under the historical hydroclimate; the management effect is this strategy minus current operations under that same hydroclimate. Axis convention [1, 4.99] per `Ordinal_vs_continuous_tiers.md`; the visible band is `floor(value)`. A difference of 0.1 on the level scale is required before a difference is reported as movement, before a location is counted as differing from its comparison, and before an effect prints as anything other than 0.0. Color semantics: warm = hydroclimate, blue = management. Strategy identity is its descriptive name; run IDs appear only in this note and the data file.

Explore further

Every value on this page comes from the outcome-level data, and the data explorer holds the same data without this brief's selection of it. The explorer holds the full scenario set and the time series behind the outcome levels shown here. A link that opens WBA9 - Butte City directly is not yet available: the per-location address scheme for the explorer is still being settled, and this brief does not guess at one.

Open the COEQWAL data explorer.

How to read the values

The level scale runs from 1 to 4.99. Level 1 is the favorable end of the scale: the outcome-level framework names Level 1 Optimal and Level 4 Critical, so a smaller value sits nearer Optimal and a larger value nearer Critical. The API serves this number in a field named `weighted_score`; that name describes how the number is computed, and does not mean that a larger number is a more favorable result.

Two numbers describe each result. The integer `tier_level` is the level label. `tier_continuous` is the position within the band of that label, and always falls inside it: `tier_continuous` is at least `tier_level` and less than `tier_level` plus one. Labels, counts and prose use `tier_level`; figure positions and statistics use `tier_continuous`.

Level 1 is Optimal, Level 2 is Acceptable, Level 3 is At-risk, and Level 4 is Critical. These are definitional category names supplied by the outcome-level framework; what a category means for any reader, community, or decision is not specified by COEQWAL.

A key outcome value is the unweighted mean over the locations in that key outcome. Every location counts equally, regardless of the people, acres, or acre-feet it represents.

Any difference between two values under different hydroclimates contains both a hydroclimate effect and a management effect; the two are separable only through the decomposition, because each is defined against its own reference. A strategy is compared with current operations under the same hydroclimate. Comparing one strategy with itself across hydroclimates is a different quantity: the hydroclimate effect plus the change in the management effect. The hydroclimate effect is always measured on current operations, not on the strategy.

One threshold governs every claim and every count here. A difference smaller than 0.1 on the level scale is not reported as movement, and a location is not counted as differing from its comparison. An outcome level is a summary built so that results in nine key outcomes with no shared unit can be compared, and 0.1 on the level scale is the finest reading it carries; a smaller difference is left unstated rather than described.

The five hydroclimates were chosen to span a range of climate stress levels. They are not a sequence, a trend, or a forecast, they carry no probabilities, and no mean is taken over them.

A run ID identifies one modeling run of one strategy under one hydroclimate. It is an internal identifier, used for citation and for tracing a value back to the data platform, not a name for the strategy; the public identity of a strategy is its descriptive name.

Level values are reported to one decimal place throughout this brief — in the prose, in the figure and in the table. The level scale does not carry meaning finer than that. Each effect shown is the difference between the floored values, except that an effect smaller than 0.1 on the level scale, which is the threshold this brief reports movement at, is shown as 0.0, so a row containing one does not add up. Values are floored rather than rounded, so a printed value stays inside the band its own label names; full-precision values are in the machine-readable sidecar this brief is built from.